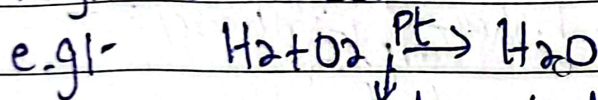


• Factors which effect rate of reaction:- (1st lecture ^{termining} part)

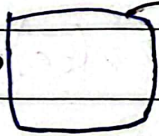
3- Temperature:- when ~~rate~~ temperature increases then the rate of temperature increases and when temperature decrease then rate ^(speed) of temperature decreases. It is roughly estimated that for every 10°C rise in temperature ^{then} rate of reaction doubles.

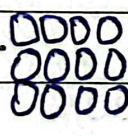
4- presence of catalyst $A + B \rightarrow P$
~~that a substance that~~ ^{chemical} ~~that~~ ^{cat (platinum) metal} increases the rate of reaction in reactant but ~~itself~~ cannot chemically used catalyst itself (not consumed).
 chemical change $\rightarrow$ not reverse
 physical change $\rightarrow$ can be reverse (e.g. color can be change when catalyst increases the rate of reaction chemical change cannot occur but physical change may or maynot be occur.)



platinum (catalyst) increases rate of reaction.

5- Surface Area:-

coal bulk $\rightarrow$  surface area upper wali hota (outer) inner wali surface area nhi hota.

coal powder $\leftarrow$  $\rightarrow$ surface area iska zada

• surface area high $\xrightarrow{\text{then}}$ chemical reaction fast (rate of reaction fast because large surface area)

• surface area smaller (bulk) $\rightarrow$ chemical reaction ~~slow~~ slow.

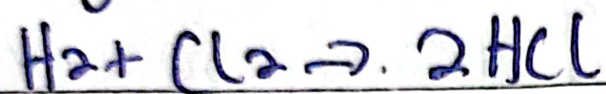
e.g. bulk of coal ~~can't~~ catches fire late as compare to coal (left) in powder form which catches fire much fast.

Day: _____

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6- Radiations: The rate of reaction increases when exposed to radiation (sunlight)

e.g. when hydrogen and chlorine reacts with the presence of sunlight then rate of reaction increases.



- Factors affecting rate of reaction.

• Concentration.

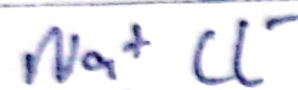
• When reactant concentration is higher than chances of collision is higher then rate of reaction is higher.

- Nature of reaction.

fastest rate

→ Ionic nature

→ Covalent nature



1 step

Sugar

↓
Yeh molecules

form ma hi ruke ga.

is ma jb mix hoo ga yeh tb phle break hoga phir reaction ho ke bond formation hoga.

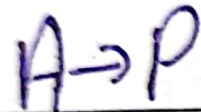
2 steps

↓
bond breaking +
bond formation

is ma water ke sath dissolve hoga to Na alag hoga Cl alag Now separate hoo ke baad reaction start hoga. to reaction jb start hoga to bond formation start hogi.

reaction jb start hoga to bond formation start hogi.

Kinetics of 1st Order.



when $t=0$ a moles dm^{-3}

0 moles dm^{-3}

when $t=t$ $(a-x)$ moles dm^{-3}

x moles dm^{-3}

• According to law of mass action,
rate $\propto (A)^1$

$$\frac{du}{dt} \propto (a-u)^1$$

$$\frac{du}{dt} = k(a-u)$$

∴ Separate concentration and other things as du is change in concentration and $(a-u)$ is also concentration.

$$\frac{du}{(a-u)} = k dt \rightarrow (1)$$

Now taking integration of eq (1)

$$\int \frac{du}{(a-u)} = \int k dt$$

$$\int \frac{du}{(a-u)} = k \int dt$$

from eq (4)

$$-\ln(a-u) = kt + C \rightarrow (2)$$

By applying boundary or initial condition

$$-\ln(a-u) = kt + C$$

when $t=0$ and $u=0$

$$-\ln(a-0) = k(0) + C$$

$$-\ln a = C$$

$$C = -\ln a \rightarrow (3)$$

putting C value in eq (2)

$$-\ln(a-u) = kt - \ln a$$

$$\ln a - \ln(a-u) = kt$$

$$kt = \ln \frac{a}{a-u} \quad \therefore \ln a - \ln b = \ln \frac{a}{b}$$

$$k = \frac{1}{t} \ln \frac{a}{a-u}$$

• To solve $\int \frac{du}{(a-u)} \rightarrow (1)$

• Suppose that $(a-u) = y$
 $(a-u) = y \rightarrow (2)$

• Now taking differentiation on both sides.

$$d(a-u) = dy$$

$$da - du = dy$$

$$0 - du = dy$$

$$du = -dy \rightarrow (3)$$

putting eq (2) and (3) in (1)

$$\int \frac{du}{(a-u)} = - \int \frac{dy}{y}$$

$$= -\ln y$$

from eq (2) as $y = (a-u)$

$$= -\ln(a-u) \rightarrow (4)$$

∴ converting natural log to normal log.

↑ natural log
↑ common log
∴ $\ln = 2.303 \log$

$$k = \frac{2.303}{t} \log \frac{a}{(a-u)}$$

Unit of 1st order

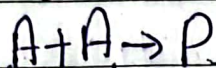
$$k = \frac{2.303}{t} \log \frac{a}{(a-u)} \rightarrow \frac{\text{mol dm}^{-3}}{\text{mol dm}^{-3}} \rightarrow \text{mol dm}^{-3}$$

$$k = \frac{1}{\text{sec}} \frac{\text{mol dm}^{-3}}{\text{mol dm}^{-3}} \quad \therefore \log = \text{constant}$$

$$k = \text{sec}^{-1}$$

Kinetics of 2nd Order.

Case-1- when concentration of both reactants are same



when $t=0$ $a \text{ mol dm}^{-3}$ $a \text{ mol dm}^{-3}$ 0 mol dm^{-3}

when $t=t$ $(a-u) \text{ mol dm}^{-3}$ $(a-u) \text{ mol dm}^{-3}$ $u \text{ mol dm}^{-3}$

∴ According to law of mass action

rate $\propto (a-u)(a-u)$

$$\frac{du}{dt} \propto (a-u)^2$$

$$\frac{du}{dt} = k(a-u)^2$$

∴ Separating concentration terms

$$\frac{du}{(a-u)^2} = k dt \rightarrow (1)$$

Now taking integration of eq (1).

$$\int \frac{du}{(a-u)^2} = \int k dt$$

$$\int \frac{du}{(a-u)^2} = k \int dt \rightarrow (2)$$

Date: _____

putting eq (4) in eq (2).

$$\frac{1}{(a-u)} = kt + C \rightarrow (3)$$

By applying boundary or initial condition on eq (3)

when $u=0$, $t=0$

$$\frac{1}{(a-0)} = k(0) + C$$

$$C = \frac{1}{a} \rightarrow (4)$$

put C value in eq (3)

$$\frac{1}{(a-u)} = kt + \frac{1}{a}$$

$$\frac{1}{(a-u)} - \frac{1}{a} = kt$$

$$\frac{a - (a-u)}{a(a-u)} = kt$$

$$\frac{u}{a(a-u)} = kt$$

$$\frac{u}{a(a-u)} = kt$$

$$k = \frac{1}{t} \cdot \frac{u}{a(a-u)}$$

Unit of 2nd order

$$k = \frac{1}{\text{Sec}} \cdot \frac{\text{mol dm}^{-3}}{\text{mol dm}^{-3} (\text{mol dm}^{-3})}$$

$$k = \text{Sec}^{-1} \text{ mol}^{-1} \text{ dm}^3$$

$$k = \text{dm}^3 \text{ moles}^{-1} \text{ Sec}^{-1}$$

To solve $\int \frac{du}{(a-u)^2}$

$\therefore$ Suppose $(a-u) = y$

$$d(a-u) = dy$$

$$da - du = dy$$

$$0 - du = dy$$

$$du = -dy \rightarrow (3)$$

$$\int \frac{du}{(a-u)^2} = - \int \frac{dy}{y^2} = - \int y^{-2} dy$$

$$= \frac{-y^{-2+1}}{-2+1}$$

$$= \frac{-y^{-1}}{-1}$$

$$= \frac{1}{y} = \frac{1}{(a-u)}$$

$$k = \text{sec}^{-1} \text{mol}^{-1} \text{dm}^3$$

So $k = \text{dm}^3 \text{mol}^{-1} \text{sec}^{-1}$

Dependence of Temperature on rate of reaction.

$$k = A e^{-E_a/RT}$$

$\therefore$ Arrhenius equation

- Exponential form of Arrhenius reaction. no derivation, equation derived from direct result.
- as there is neg sign rate constant now directly proportional to temperature so E_a (activation energy) increases temperature decreases.

k = rate constant.

A = Arrhenius constant / pre exponential factor / frequency factor

$e^{-E_a/RT}$ = Boltzmann factor

E_a = activation energy

R = gas constant

T = temperature in kelvin (absolute temperature)

$$k = A e^{-E_a/RT} \rightarrow (1)$$

Simple form of exponential form of Arrhenius reaction

Taking log on both sides

$$\ln k = \ln A e^{-E_a/RT}$$

$$\ln k = \ln A + \ln e^{-E_a/RT}$$

$$\therefore \ln AB = \ln A + \ln B$$

$$\ln k = \ln A - \frac{E_a}{RT} \ln e$$

$\therefore$ As $\ln e = 1$

$$\ln k = \ln A - \frac{E_a}{RT} \rightarrow (2)$$

Logarithmic form

Converting to common log

$$\therefore \ln x = 2.303 \log x$$

$$2.303 \log k = 2.303 \log A - \frac{E_a}{RT}$$

Dividing 2.303 on both sides

$$\frac{2.303}{2.303} \log k = \frac{2.303}{2.303} \log A - \frac{E_a}{2.303 RT}$$

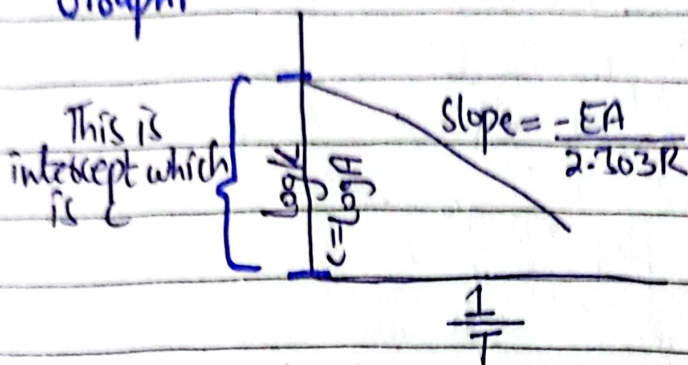
$$\log k = \log A - \frac{E_a}{2.303RT} \rightarrow (3)$$

From straight line equation

c = intercept

∴ as m negative so curve downward m (slope)

Graph



$$\text{So } E_a = -\text{Slope} \times 2.303R$$

∴ Why we take log?

$$16 = 2^4 \rightarrow \text{power}$$

$$\log_{\text{base}} 16 = 4$$

∴ from log we came for multiply to add and divide to subtract

∴ Differential form of eq (2)

∴ differential of k with respect to temperature.

$$\frac{d}{dT} \ln k = \frac{d}{dT} \ln A - \frac{d}{dT} \left(\frac{E_a}{RT} \right)$$

$$\frac{d}{dT} \ln k = 0 - \frac{E_a}{R} \frac{d}{dT} T^{-1}$$

$$\frac{d}{dT} \ln k = -\frac{E_a}{R} T^{-1-1}$$

$$\frac{d}{dT} \ln k = \frac{E_a}{R} T^{-2}$$

$$\frac{d}{dT} \ln k = \frac{E_a}{RT^2}$$

∴ a = arrhenius constant
So it does not depend on any temperature so with differential it becomes 0.

∴ k is also constant but on specific temperature not on all temperature

$$\boxed{\frac{d \ln k}{dT} = \frac{E_a}{RT^2}} \rightarrow (4)$$

∴ just separate temp to one side.

∴ Differential form of arrhenius reaction.

Day:

within limit

Date:

Integration without limit

Integration within limit of eq. (4)

 T_1 = initial temperature k_1 = initial rate constant T_2 = final temperature k_2 = final rate constant

$$\int_{k_1}^{k_2} d \ln k = \int_{T_1}^{T_2} \frac{E_a}{RT^2} dt$$

$$\int_{k_1}^{k_2} d \ln k = \frac{E_a}{R} \int_{T_1}^{T_2} T^{-2} dt$$

$$[\ln k]_{k_1}^{k_2} = \frac{E_a}{R} \left[\frac{T^{-2+1}}{-2+1} \right]_{T_1}^{T_2}$$

$$\ln k_2 - \ln k_1 = \frac{E_a}{R} \left[\frac{-1}{T} \right]_{T_1}^{T_2} \quad \therefore T^{-1} = \frac{1}{T}$$

$$\ln k_2 = \frac{E_a}{R} \left[\frac{-1}{T_2} - \left(\frac{-1}{T_1} \right) \right] \quad \therefore \ln a - \ln b = \ln \frac{a}{b}$$

$$\ln k_2 = \frac{E_a}{R} \left[\frac{-1}{T_2} + \frac{1}{T_1} \right]$$

$$\ln k_2 = \frac{E_a}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\ln k_2 = \frac{E_a}{R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

Now converting to common log

$$2.303 \log \frac{k_2}{k_1} = \frac{E_a}{R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_1 T_2} \right] \rightarrow (5)$$

Date: 4th Nov, 24

Chemistry Theory lecture 6. Imp.

form of energy

Spectroscopy

electromagnetic radiation and matter } study of their interaction.
with the help of spectroscopy we can find atom and molecule structure. Speed of light / electromagnetic radiation = $3 \times 10^8 \text{ ms}^{-1}$ or 18600 miles s^{-1}

Electromagnetic radiation 2 components

electric field

magnetic field

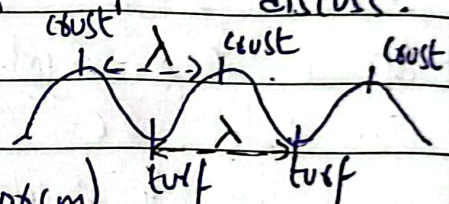
• both are coplanar and are perpendicular to each other. direction of oscillation (perpendicular)

• its direction may be propagate (move) like us' propagation ke sath hi yeh perpendicular hote

• Properties of "electromagnetic radiations" - representation, ^{def} unit, mathematical formulae, all properties we discuss.

1- Wavelength - lambda (λ) ~~unit~~ ← symbol

def - The distance between two crest or trough of a particular wave is called wavelength.

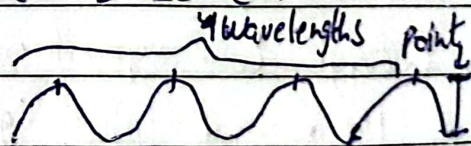


Unit - Angstrom ($\text{\AA}$) = 10^{-10} m (m or cm) $\therefore 1 \text{ m} = 100 \text{ cm}$
 $= 10^{-10} \times 10^2 \text{ cm} = 10^{-8} \text{ cm}$

Nanometre (nm) = $10^{-9} \text{ m} = 10^{-9} \times 10^2 \text{ cm} = 10^{-7} \text{ cm}$

micrometre (μ) = $10^{-6} \text{ m} = 10^{-6} \times 10^2 \text{ cm} = 10^{-4} \text{ cm}$

millionth of metre



2- Frequency - ν (ν) ← symbol

def - The number of wavelength which are passes through the point in 1 sec or unit time is called frequency.

• long wavelength then less wave passes from point so less frequency.

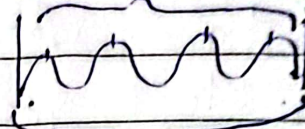
$$\nu \propto \frac{1}{\lambda} \rightarrow \text{wavelength}$$

$$\nu = \frac{c}{\lambda}$$

$$c = 3 \times 10^8 \text{ ms}^{-1}$$

Unit: Hertz (Hz) or cycle per second (s^{-1})

3- Wave Number / Number of wavelength in 1cm distance is their wavenumber.



1cm distance

nubas ($\bar{\nu}$) ← "symbol"

• if wavelength long then less wavelength in 1cm distance so less wave numbers.

$$\bar{\nu} \propto \frac{1}{\lambda}$$

wavenumber $\leftarrow \bar{\nu} = \frac{1}{\lambda \rightarrow \text{wavelength}}$ $\therefore$ here proportional constant is 1 not c.

$$\text{Unit: } \bar{\nu} = \frac{1}{\lambda} = \frac{1}{\text{cm}} = \text{cm}^{-1}$$

same $\rightarrow$ velocity $\rightarrow$ change

4- Speed: $c \leftarrow$ symbol of representation

$$\text{Speed of light} \rightarrow c = 3 \times 10^8 \text{ ms}^{-1}$$

• Change of frequency, wavelength, wavenumber does not have any effect on speed of light / electromagnetic radiation and their constant value will be always $3 \times 10^8 \text{ ms}^{-1}$.

Unit: ms^{-1}

5- Energy: $E \rightarrow$ symbol of representation

$\therefore$ According to plank quantum theory

$$E \propto \nu \rightarrow \text{frequency (or } \nu)$$

Energy of electromagnetic radiation

$$E = h \nu$$

electromagnetic radiation

$\therefore h = \text{plank constant}$

$$h = 6.626 \times 10^{-34} \text{ Js}$$

Unit: Js

frequency

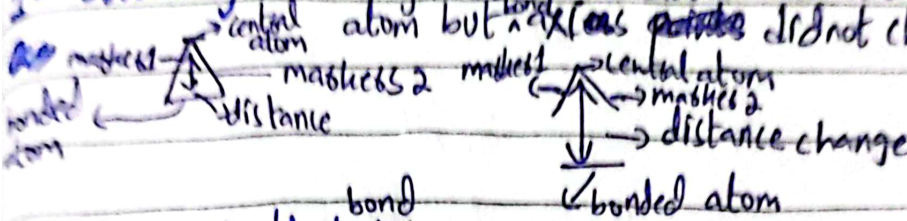
Imp.

Chemistry Theory Lecture 6.

Types of Vibration - When distance between two atoms increase or decrease then it is called vibration.

2 Types

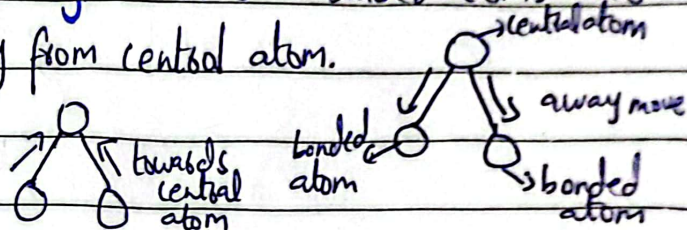
1- **Stretching vibration** - of bond distance changes between central atom and bonded atom but ~~angle~~ ~~points~~ ~~points~~ did not change.



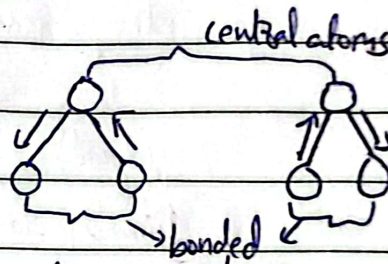
2- **Bending vibration** - distance remains same but bond ~~axis~~ ~~axis~~ changes.

Stretching vibration types

1- **Symmetric stretching** - Either both bonded atoms move towards or away from central atom.



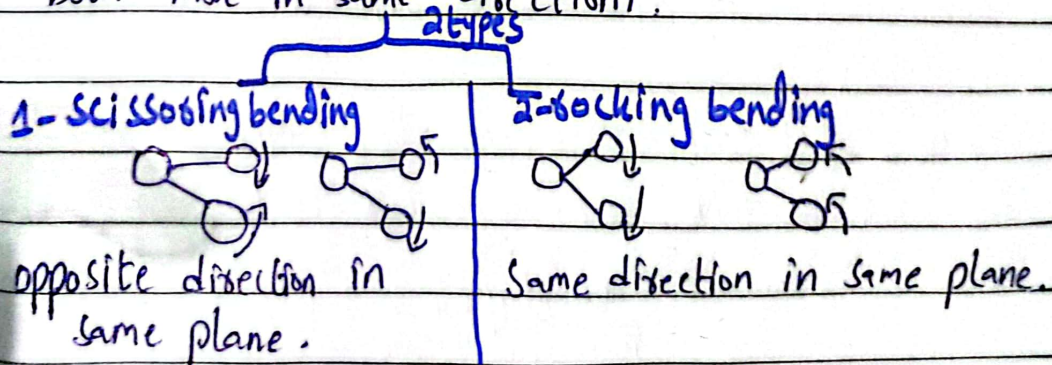
2- **Asymmetric stretching**



One may move towards central atom while others may move away from central atom.

Bending vibration types

1- **Inplane bending** - plane does not change (bonded atoms both move in same direction).

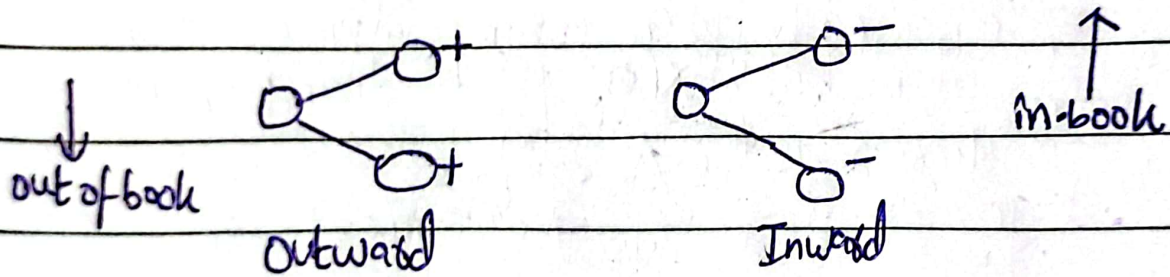


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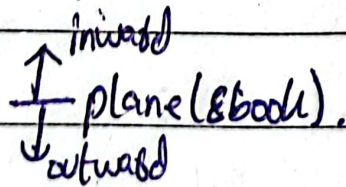
Date: _____

2- Outplane bending:- plane does change lya inward ja baha ya outward ja baha).

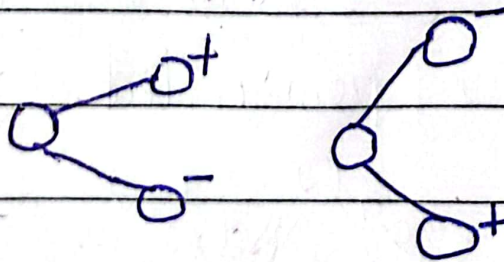
1- Wagging bending:-



Both atom have same direction. Both go in the same direction either into the plane or away from the plane.



2- Twisting bending:-



Both atoms go in the opposite direction from the plane.